

PRAGYA SINGHAL

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SUMMARY

Software engineer, ~4 years building production systems. Now in GenAI - I design & ship agentic AI on FastAPI: a stateful LangGraph tool-calling agent (6 tools, structured JSON outputs, HITL safety gates, error recovery with retry-with-backoff and circuit breakers) and a grounded RAG system with automated evaluation harnesses (hallucination checks, retrieval quality, latency/cost tracking). Strong in enterprise REST API integration, design patterns, and responsible-AI guardrails.

GENAI PROJECTS

MyHive — Production Conversational Assistant

[myHive](#)

OpenAI GPT, Firebase, Ionic, Angular, TypeScript, Node.js

- Shipped a production conversational assistant (OpenAI GPT) on MyHive, a cohort-based wellness marketplace-answers users' health queries and recommends available practitioners on the platform.
- Built personalized exercise recommendations that map each user's mood, pain, sleep, and daily health cycles to relevant programs from the platform's catalog.

HomeoQuiz — Grounded RAG with Faithfulness Evaluation

[homeoQuiz](#)

Python · Gemini 2.5 Flash-Lite · ChromaDB · sentence-transformers · HF Spaces

- Built a strict-grounding RAG question-answering system over a fixed canonical corpus where every answer and explanation cites a source passage, an automated eval harness (Wilson 95% CIs) held hallucination_rate at 0.0 and quote_validity at 1.0 across 102 served queries.
- Added a guardrail that refuses to answer when the source material doesn't cover a topic - 100% correct refusals (27/27), while rarely over-refusing (2.9%).
- Engineered a hybrid local-embed (bge-small) + remote-generate(Gemini 2.5 Flash-Lite) architecture - a deliberate cost/latency/rate-limit tradeoff - reaching retrieval_hit@k 0.87 (n=30) at p50 ~2.0s latency.

CRM Agent — Stateful Tool-Calling Agent with Safety & Eval

[crmAgent](#)

LangGraph · FastAPI · Supabase · Gradio · HF Spaces

- Built and deployed a stateful, tool-calling CRM agent having 6 tools with structured outputs, human-in-the-loop approval gates(prompt independent), error recovery (retry-with-backoff + circuit breaker), and cost tracking.
- Made risky actions safe by design- a code-level approval gate catches every sensitive action that reaches it and guardrails raised how many sensitive actions reach the gate from 9/14 to 13/14.
- Tested it with a 50-query eval harness comparing guardrails off vs on (3 runs each, fully stable) - guardrails lifted tool-selection and parameter-extraction to 100%.

PROFESSIONAL EXPERIENCE

Software Engineer | 29 Kreativ

Aug 2022 - Present

- Integrated OpenAI GPT in production (Chatbot & AI-assisted features), the team's first user-facing LLM capabilities.
- Built event-driven workflows for CRM operations, payment-status synchronization, and customer notifications across a platform serving ~40K users, reducing manual operational effort by ~75–80%.
- Designed and shipped REST APIs and integrated external enterprise systems through APIs and webhooks, including Stripe and Secuconnect, handling authentication, data mapping, error handling, and production debugging.
- Led a legacy CodeIgniter 3 → 4 migration and a DataTables.js → Tabulator.js migration with zero regression on a large Laravel application, designed and shipped REST APIs across products.
- Mentored 5+ junior developers on engineering practices, debugging, and cross-stack troubleshooting.

TECHNICAL SKILLS

GenAI/LLM: RAG, agentic AI, prompt engineering, tool calling & structured outputs, workflow orchestration, MCP, vector DBs (ChromaDB, FAISS), embeddings, LLM evaluation, guardrails, cost/latency optimization

LLM APIs & Frameworks: LangChain, LangGraph(agent state machines, HITL interrupts), FastAPI, OpenAI, Gemini, Groq

Backend & Languages: TypeScript, Python, Node.js, PHP, Laravel, CodeIgniter, REST API design

Databases & Deploy: MySQL, NoSQL (Firebase, ChromaDB), Docker, Hugging Face Spaces, Git/GitHub, CI/CD

EDUCATION

B.Tech in Electronics & Communication Engineering – ADGITM, Delhi

2018 - 2022